Documenting a Synthetic Sales Display Script

Mahmoud Bakar

6 November 2026

AI Assistance Declaration: ChatGPT (GPT-6 family, 6 November 2026) assisted with documentation structure, wording, formatting, illustrative R code and reflection drafts. Prompts are recorded in Appendix.md. The assistant checked the Markdown, ran the R script and knitted the R Markdown document. This example contains no analytical calculations. Student review is required before signing the instructor's personal declaration.

1 Purpose

This document explains `sales_display.R`, a small base-R script that creates and prints fictional monthly sales observations. It documents a single script without claiming statistical findings. The example demonstrates reproducible documentation rather than an analytical calculation.

2 Inputs

The script defines its data internally. It does not download records, read external files or accept command-line arguments.

Field	R type	Meaning
Month	character	Fictional observation month
Product	character	Fictional item label
Units	integer	Fixed example quantity; not a calculated value

The three months are January, February and March. The product is Notebook. The fixed unit values are 12, 15 and 9. No personal information is present.

3 Outputs

The script creates an in-memory data frame named `sales` and prints it without row numbers. It creates no persistent data file and changes no input file. Knitting this Rmd creates an HTML documentation file separately.

4 Requirements and Execution

The script needs only base R. Knitting needs `rmarkdown`, `knitr` and Pandoc. Set the working directory to the repository root when running the example manually.

4.1 Run the Documented Script

This chunk executes the script itself so that the documentation and implementation stay connected. R Markdown normally evaluates relative paths from the document directory, where the script is located.

```
source("sales_display.R", local = knitr::knit_global())
```

```
##     Month  Product Units
##   January Notebook    12
##  February Notebook    15
##     March Notebook     9
```

4.2 Script Walkthrough

`data.frame()` combines the named vectors into a table. The `L` suffix makes the unit constants integers. `stringsAsFactors = FALSE` keeps labels as character values. `print()` sends the table to the console; `row.names = FALSE` hides row numbers. The script contains no arithmetic or aggregation.

5 Validation

The expected rendered output contains the same three fictional records as the README. Compare each field against the script rather than trusting generated prose. Confirm that the headings, input table and R code appear correctly after knitting.

The assistant executed the script successfully and knitted this document using R 4.3.3, rmarkdown 2.25, knitr 1.45 and Pandoc 3.1.3. The printed rows were compared with the fixed input values and README table. The HTML output includes the actual execution result. This technical check does not represent a personal student review.

6 Limitations and Responsible Use

The example does not handle imported datasets, missing values or real sales analysis. It is intentionally small so documentation can be inspected directly. AI assistance with prose and illustrative code is disclosed. The student remains responsible for reviewing and understanding the submission.

7 Related Files

See README, AI Appendix and Verification for project setup, attribution and pending checks.